



Proposed Solution Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

Solution Overview

Attribute	Details
Solution Name	Rising Waters – ML-Based Flood Prediction System
Approach	Supervised Machine Learning — Binary Classification (Flood / No Flood)
Dataset	5,000 synthetic meteorological records; 6 input features; binary target
Algorithms Tested	Decision Tree (78.17%), KNN (80.67%), Random Forest (82.00%), XGBoost (96.55%)
Best Model	XGBoost — 96.55% accuracy; saved as floods.save via Joblib
Web Application	Flask app with 6 HTML pages including dedicated result pages
Deployment	Docker containerized → IBM Cloud Foundry
Key Innovation	Actionable emergency protocols based on prediction outcome

Model Performance Comparison

S.No	Algorithm	Train Accuracy	Test Accuracy	Deployed
1	Decision Tree	82.40%	78.17%	No
2	K-Nearest Neighbours	85.50%	80.67%	No

S.No	Algorithm	Train Accuracy	Test Accuracy	Deployed
3	Random Forest	88.00%	82.00%	No
4	XGBoost	99.20%	96.55%	Yes